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ELECTRONIC DEVICE

Abstract

An electronic device comprising semiconductor-based power transistors is provided. An example electronic device comprises a transistor. The transistor comprises: a p-doped semiconductor first layer; a piezoelectric semiconductor second layer, covering the first layer; a piezoelectric semiconductor third layer, covering the second layer; a piezoelectric semiconductor fourth layer, covering the third layer; and a piezoelectric semiconductor fifth layer, covering the fourth layer, the transistor being configured to generate a first two-dimensional electron gas between the fourth and fifth layers; and a gate pattern crossing going through the fifth layer and at least part of the fourth layer.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims the priority benefit of French patent application number FR2402000, filed on Feb. 29, 2024, entitled "Dispositif électronique", which is hereby incorporated by reference to the maximum extent allowable by law.

TECHNICAL FIELD

[0002] The present disclosure relates generally to electronic devices and more especially electronic devices comprising semiconductor-based power transistors.

BACKGROUND

[0003] In the field of power electronics, a significant challenge relates to the development of "large band gap" semiconductor-based power transistors, such as GaN.

[0004] Different architectures have been specifically developed for these GaN-based transistors. The architectures based on the use of a two-dimensional electron gas (2 DEG), typically making it possible to operate with high source-drain biasing voltages. A positive threshold voltage advantageously makes it possible to simplify the control circuit of the transistor and to ensure the safety of the conversion system in the event of a failure.

BRIEF SUMMARY

[0005] One embodiment provides a device comprising a transistor, the transistor comprising: —a p-doped semiconductor first layer; —a piezoelectric semiconductor second layer, covering the first layer; —a piezoelectric semiconductor third layer, covering the second layer; —a piezoelectric semiconductor fourth layer, covering the third layer; —a piezoelectric semiconductor fifth layer, covering the fourth layer, the transistor being configured to generate a first two-dimensional electron gas between the fourth and fifth layers; and —a gate pattern crossing going through the fifth layer and at least part of the fourth layer.

[0006] According to an embodiment, the transistor is configured to generate a second two-dimensional electron gas between the second and third layers, the first layer being configured to suppress the second electron gas.

[0007] According to an embodiment, the gate pattern crosses the third, fourth and fifth layers and at least part of the second layer.

[0008] According to an embodiment, the second and fourth layers are made of the same material.

[0009] According to an embodiment, the second and fourth layers are made of GaN.

[0010] According to an embodiment, the third and fifth layers are made of the same material.

[0011] According to an embodiment, the third and fifth layers are made of AlGaN, AsGa, AlN or InGaN.

[0012] According to an embodiment, the first layer is made of GaN doped with magnesium, carbon or iron.

[0013] According to an embodiment, the first layer is configured not to be biased.

[0014] According to an embodiment, the gate pattern separates at least the fifth layer in a first part and a second part.

[0015] According to an embodiment, the gate pattern separates the third, fourth and fifth layers in a first part and a second part.

[0016] According to an embodiment, the device comprises a first electrode in contact with the first part of fifth layer and a second electrode in contact with the second part of fifth layer.

[0017] According to an embodiment, the first layer is configured to be biased with the same voltage as either the first or second electrode.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0018] The foregoing features and advantages, as well as others, will be described in detail in the following description of specific embodiments given by way of illustration and not limitation with reference to the accompanying drawings, in which:

[0019] FIG. **1** illustrates an example of an electronic device comprising a semiconductor-based power transistor;

[0020] FIG. **2** illustrates an embodiment of an electronic device comprising a semiconductor-based power transistor;

[0021] FIG. **3** illustrates the behavior of the electronic device of FIG. **2**; and

[0022] FIG. **4** illustrates another embodiment of an electronic device comprising a semiconductor-based power transistor.

DETAILED DESCRIPTION

[0023] Like features have been designated by like references in the various figures. In particular, the structural and/or functional features that are common among the various embodiments may have the same references and may dispose identical structural, dimensional and material properties.

[0024] For the sake of clarity, only the operations and elements that are useful for an understanding of the embodiments described herein have been illustrated and described in detail.

[0025] Unless indicated otherwise, when reference is made to two elements connected together, this signifies direct connection without any intermediate elements other than conductors, and when reference is made to two elements coupled together, this signifies that these two elements can be connected or they can be coupled via one or more other elements.

[0026] In the following disclosure, unless indicated otherwise, when reference: positional qualifiers, such as the terms “front”, “back”, “top”, “bottom”, “left”, “right”, etc., or to relative positional qualifiers, such as the terms “above”, “below”, “higher”, “lower”, etc., or to qualifiers of orientation, such as “horizontal”, “vertical”, etc., reference is made to the orientation shown in the figures.

[0027] Unless specified otherwise, the expressions “around”, “approximately”, “substantially” and “in the order of” signify within 10%, and preferably within 5%.

[0028] FIG. **1** illustrates an example of an electronic device comprising a semiconductor-based power transistor.

[0029] More precisely, FIG. **1** illustrates schematically a part of a device comprising the power transistor **10**. The transistor **10** is a High Electron Mobility Transistor (HEMT)-type transistor. Such a transistor is a field effect transistor with high electron mobility, also sometimes referenced by the term of heterostructure field effect transistor.

[0030] The transistor **10** comprises a stack **12** of layers **14**, **16**, **18**, **20**, **22**. The stack **12** is for example located in or on a semiconductor substrate **13**. The substrate **13** is for example made of silicon, sapphire or SiC. In the example of FIG. **1**, the stack **12** is located on the upper face of the substrate **13**.

[0031] The layer **14** is a buffer layer. The layer **14** can correspond to several buffer layers. The buffer layer **14** is for example in AlN, GaN or AlGaN.

[0032] The layer **16** is in a piezoelectric material, for example in GaN. The layer **16** is for example intrinsic or unintentionally doped. In other words, layer **16** is either not doped or doped with a doping concentration below 1014 cm-3. Layer **16** is located on, and for example in contact with, layer **14**. The layer **16** is for example floating. In other words, layer **16** is for example not biased. Alternatively, layer **16** is biased by the reference voltage, for example the ground.

[0033] The layer **18**, which constitutes a back barrier, is in a piezoelectric material, for example in AlGaN. The proportion of aluminum is for example higher than 4%. Layer **18** is located on, and for example in contact with, layer **16**. The layer **18** is for example floating.

[0034] The layer **20**, which constitutes a channel, is in a piezoelectric material, for example in GaN. Layer **20** is located on, and for example in contact with, layer **18**.

[0035] The layer **22**, which constitutes a barrier, is in a piezoelectric material, for example in AlGaN. The proportion of aluminum is for example higher than 4%. Layer **22** is located on, and for example in contact with, layer **20**. Layer **22** for example does not cover entirely layer **20**. In other words, a part of layer **20**, for example a part of the upper face of layer **20**, is not covered by the layer **22**.

[0036] The transistor **10** comprises a gate pattern **24**. The gate pattern **24** is located in a cavity **26**. The cavity extends from the upper face of layer **22**, through layer **22** and part of layer **20**. In other words, the cavity **26** extends through the interface between layers **20** and **22**. The cavity **26** extends through layers **18**, **20**, **22** and through part of layer **16**. In other words, the bottom of the cavity **26** of FIG. **1** is located in the layer **16**.

[0037] The layers crossed by the cavity, for example the layers **18**, **20**, **22** in FIG. **1**, are divided in two parts by the cavity **26**. Layer **18** is divided in parts **18***a* and **18***b*, parts **18***a* and **18***b* not being in contact with each other. Layer **20** is divided in parts **20***a* and **20***b*, parts **20***a* and **20***b* not being in contact with each other. Layer **22** is divided in parts **22***a* and **22***b*, parts **22***a* and **22***b* not being in contact with each other. Parts **18***a*, **20***a*, **22***a* form a secondary stack **12***a* and parts **18***b*, **20***b* and **22***b* form a secondary stack **12***b*. The secondary stacks **12***a* and **12***b* are separated by the cavity **26**.

[0038] The layer **16** is a continuous layer. In other words, the layer **16** is made of a single part. The portion of layer **16** facing parts **18***a*, **20***a*, **22***a*, for example in contact with part **18***a*, and the portion of layer **16** facing parts **18***b*, **20***b*, **22***b*, for example in contact with part **18***b*, are in contact with each other. Said portions of layer **16** are, therefore, not separated by a layer in a different material.

[0039] The gate pattern **24** comprises a dielectric layer **28** and a conductive layer **30**. Layer **28** covers conformally the cavity **28** and part of the upper face of the layer **22** surrounding the cavity **26**. In other words, layer **28** covers the lateral face of the layers **16**, **18**, **20**, **22** forming the walls of the cavity, the face of the layer **16** forming the bottom of the cavity **26** and part of the upper face of the layer **22** surrounding the cavity **26**. Layer **30** covers conformally layer **28**. Preferably, layer **30** covers entirely, and only, layer **28**. Layer **30** is therefore separated from stack **12** by layer **28**.

[0040] The transistor **10** comprises a drain electrode **32** and a source electrode **34**. The drain electrode **32** is in contact with both layers **20** and **22** on one side of the cavity **26** and the source electrode **34** is in contact with both layers **20** and **22** on the other side of the cavity **26**. More precisely, the drain electrode **32** is in contact with parts **20***a* and **22***a* of the layers **20** and **22** and the source electrode **34** is in contact with parts **20***b* and **22***b* of the layers **20** and **22**. In the example of FIG. **1**, the drain electrode **32** covers part of the upper face of part **22***a*, part of the upper face of part **20***a* which is not covered by part **22***a*, and the lateral face of part **22***a*. Similarly, the source electrode **34** covers part of the upper face of part **22***b*, part of the upper face of part **20***b* which is not covered by part **22***b*, and the lateral face of part **22***b*.

[0041] The architecture of a HEMT transistor such as transistor **10** includes the superposition of two semiconductive layers having different band gaps which form a quantum well at their interface. This quantum well is induced by the present spontaneous and piezoelectric biasing charges. Electrons are confined in this quantum well to form a two-dimensional electron gas (2DEG). In the stack **12** of FIG. **1**, a two-dimensional electron gas **36** is formed under the layer **22**, in the layer **20**. The two-dimensional electron gas **36** is typically confined at the interface between the layers **20** and **22**. Layers **20** and **22** form an heterojunction.

[0042] The gate pattern **24** passes through this interface between layers **20** and **22**, so as to interrupt the two-dimensional electron gas **36**. Controlling the gate voltage Vgs makes it possible to enable or block the passage of electrons of the two-dimensional electron gas **36** on either side of the gate pattern **24**.

[0043] Generally, if the gate of the transistor is set to a voltage greater than a threshold voltage, there is an accumulation of electrons under the gate dielectric, thus connecting the two-dimensional electron gas on either side of it, which makes it possible to connect the source and the drain such that the transistor enters an on state. Therefore, when the gate pattern **24** is in a state to enable the passage of electrons, the two-dimensional electron gas **36** extends between the parts **20***a* and **22***a*, along the walls and the bottom of the cavity **26** and between the parts **20***b* and **22***b*.

[0044] If the gate of the transistor is set at a voltage less than the threshold voltage, the source and the drain are no longer connected and the transistor enters an off state.

[0045] A second two-dimensional electron gas **38** is formed under the layer **18**, in the layer **16**. The two-dimensional electron gas **38** is typically confined at the interface between layers **16** and **18**. The two-dimensional electron gas **38** generates an unwanted constant leakage current flowing between the drain and the source of transistor **10**.

[0046] FIG. **2** illustrates an embodiment of an electronic device comprising a semiconductor-based power transistor **11**. More precisely, FIG. **2** illustrates schematically a part of a device comprising the power transistor **11**. The transistor **11** is a High Electron Mobility Transistor (HEMT)-type transistor. Such a transistor is a field effect transistor with high electron mobility, also sometimes referenced by the term of heterostructure field effect transistor.

[0047] The device is for example intended for the automotive industry. The electrification of automotive vehicles generates an expanding high level of electronic content in vehicles. The device for example comprises High Electron Mobility Transistor (HEMT) to be incorporated in the vehicles. The automatization of driving also generates an expanding high level of electronic content in vehicles.

[0048] The device can for example be used in the industrial field. More especially, the device for example aims at being used for the development of green energies or for the electrification of infrastructures, for example for charging stations or for the incorporation of solar energy. The device can also be used in the field of the internet of things and of smart homes. The device is for example intended for being implemented in the power and energy circuits of pieces of equipment, comprising for example 650V or 1200V HEMT, 1200V ultrafast and silicon carbide diodes, transient-voltage-suppression diodes, and protections against electromagnetic discharges. The device can also be used in the implementation of clouds, 5G networks, data centers and servers. The device for example comprises wide band gap materials.

[0049] The device is for example intended for being used in personal electronics, for example in the aim of increasing radio frequency content, in device of 5G connections or more generally in connected devices. The device is for example a smartphone or a part of a network of internet of things. The device is for example connected by 5G, WIFI or ultra-wide band. The device for example includes high speed interfaces, for example with advanced filtering and protection against electromagnetic discharges.

[0050] The device is for example intended for being used in communication equipment, or in computers and peripherals. For example, the device can be used in 5G infrastructure and dedicated data centers. The device comprises for example silicon carbide diodes, power Schottky transistors, protections against electromagnetic discharges and transient-voltage-suppression diodes. The device can also be used in satellites, comprising for example integrated passive devices for radio frequency applications.

[0051] The disclosed device comprises a HEMT-type transistor. HEMT-type transistors are typically used in high-frequency and high-power applications such as satellite communications, radar systems, battery chargers, computers, servers, automotive, lightning systems and photovoltaic systems, microwave amplifiers, and all equipment requiring power conversion, for example DC/DC or AC/DC or AC/AC or DC/AC power conversion. HEMT-type transistors may also be used in some specialized personal electronic devices such as high-end audio amplifiers or radio frequency (RF) transmitters. HEMT-type transistors are increasingly being used in car electrification, particularly in electric and hybrid vehicles.

[0052] The transistor **11** comprises, like the device **10** of FIG. **1**, a stack **12** of layers **14**, **16**, **18**, **20**, **22**. In the embodiment of FIG. **2**, the stack **12** further comprises a layer **15**. Preferably, the layers **14**, **15**, **16**, **18**, **20**, **22** of the stack **12** are disposed in this order from the substrate. The stack **12** is for example located in or on the semiconductor substrate **13**. The substrate is for example made of silicon, sapphire or SiC.
[0053] The layer **14** is a buffer layer. The layer **14** can correspond to several buffer layers. The buffer layer **14** is for example in AlGaN, AlN or GaN. Layer **14** is for example the lowest layer of the stack **12**. In other words, layer **14** is for example the closest layer to the substrate **13**.
[0054] The layer **15** is a semiconductor layer. The layer **15** is p-doped. For example, the layer **15** is in GaN. For example, the layer **15** is doped with magnesium, carbon or iron. Layer **15** is for example the second lowest layer of the stack **12**. Layer **15** is for example separated, preferably entirely, from the substrate by layer **14**. The doping concentration is for example higher than 1017 cm-3, for example substantially equal to 1018 cm-3.
[0055] Preferably, the layer **15** is floating. In other words, layer **15** is preferably not biased. Alternatively, layer **15** is biased by the reference voltage, for example the ground.
[0056] The layer **16** is in a piezoelectric material, preferably a piezoelectric semiconductor material. Layer **16** constitutes a buried channel or an interlayer. Layer **16** is preferably in a material configured to be formed, for example to be grown by epitaxy, on the layer **15**. The layer **16** is for example in GaN. The layer **16** is preferably intrinsic or unintentionally doped. In other words, layer **16** is either not doped or doped with a doping concentration below 1014 cm-3. For example, the layer **16** is in the same material as layer **15**.
[0057] Layer **16** is for example the third lowest layer of the stack **12**. Layer **16** is for example located on, and preferably in contact with, layer **15**. Layer **16** is for example separated, preferably entirely, from layer **14** by layer **15**.
[0058] Preferably, the layer **16** is floating. In other words, layer **16** is preferably not biased. Alternatively, layer **16** is biased by the reference voltage, for example the ground.
[0059] The layer **18** is in a piezoelectric material, preferably a piezoelectric semiconductor material. Layer **18** constitutes a back barrier. Layer **18** is preferably in a material configured to be formed, for example to be grown by epitaxy, on the layer **16**. The layer **18** is in a material different from the material of layer **16**. The layer **18** is for example in AlGaN, in InGaN, in AsGa or in AlN. Preferably, the layer **18** is in AlGaN, the proportion of aluminum being higher than 4%.
[0060] Layer **18** is for example the fourth lowest layer of the stack **12**. Layer **18** is for example located on, and preferably in contact with, layer **16**. Layer **18** is for example separated, preferably entirely, from layer **15** by layer **16**. Layer **18** is preferably not in contact with layer **15**.
[0061] Preferably, the layer **18** is floating. In other words, layer **18** is preferably not biased. Alternatively, layer **18** is biased by the reference voltage, for example the ground.
[0062] The layer **20** is in a piezoelectric material, preferably a piezoelectric semiconductor material. Layer **20** constitutes a channel. Layer **20** is preferably in a material configured to be formed, for example to be grown by epitaxy, on the layer **18**. The layer **20** is in a material different from the material of layer **18**. For example, the material of layer **20** is the same material as the material of layer **16**. Layer **20** is for example in GaN.
[0063] Layer **20** is for example the fifth lowest layer of the stack **12**. Layer **20** is for example located on, and preferably in contact with, layer **18**. Layer **20** is for example separated, preferably entirely, from layer **16** by layer **18**. Layer **18** is preferably not in contact with layers **15** and **16**.
[0064] The layer **22** is in a piezoelectric material, preferably a piezoelectric semiconductor material. Layer **22** constitutes a barrier. Layer **22** is preferably in a material configured to be formed, for example to be grown by epitaxy, on the layer **20**. The layer **22** is in a material different from the material of layer **20**. Layer **22** is for example in the same material as layer **18**. The layer **22** is for example in AlGaN, in InGaN, in AsGa or in AlN. Preferably, the layer **22** is in AlGaN, the proportion of aluminum being higher than 4%.

[0065] Layer **22** is for example the highest layer of the stack **12**, in other words, the layer the farthest from the substrate and the farthest from the buffer layer not represented. Layer **22** is for example located on, and preferably in contact with, layer **20**. Layer **22** is for example separated, preferably entirely, from layer **18** by layer **20**. Layer **22** is preferably not in contact with layers **14**, **15**, **16**, **18**. Layer **22** for example does not cover entirely layer **20**. In other words, a part of layer **20**, for example a part of the upper face of layer **20** is not covered by the layer **22**.
[0066] According to an embodiment, the layers **16**, **18**, **20**, and **22** are all in different piezoelectric materials. According to another embodiment, layers **16** and **20** are made of the same material and layers **18** and **22** are made of the same material.
[0067] Preferably, each of the layers **14**, **15**, **16**, **18**, **20**, **22** is an homogenous layer. Preferably, each of the layers **14**, **15**, **16**, **18**, **20**, **22** is made of a single material.
[0068] The layer **16** for example has a doping concentration lower that 1014 cm-3. The layer **18** has a thickness between 20 nm and 150 nm. The layer **20** for example has a doping concentration lower that 1014 cm-3. The layer **20** has a thickness for example between 50 nm and 200 nm. The layer **22** has a thickness between 20 nm and 100 nm.
[0069] The semiconductive materials of transistor **10** are for example chosen so as to have a wide energy band gap, for power handling (in particular, high voltage) and temperature handling reasons.
[0070] The transistor **11** comprises a gate pattern **24**. The gate pattern **24** is located in a cavity **26**. The cavity extends from the upper face of layer **22**. The cavity **26** extends at least through layer **22** and part of layer **20**. In other words, the cavity **26** extends through the interface between layers **20** and **22**. In the embodiment of FIG. **2**, the cavity **26** extends through layers **18**, **20**, **22** and through part of layer **16**. In other words, the bottom of the cavity **26** of FIG. **2** is located in the layer **16**. Preferably, the bottom of the cavity **26** is located between the interface of layers **20** and **22** and the layer **16**.
[0071] The layers crossed by the cavity, for example the layers **18**, **20**, **22** in FIG. **2**, are divided in two parts by the cavity **26**. Layer **18** is divided in parts **18***a* and **18***b*, parts **18***a* and **18***b* not being in contact with each other. Layer **20** is divided in parts **20***a* and **20***b*, parts **20***a* and **20***b* not being in contact with each other. Layer **22** is divided in parts **22***a* and **22***b*, parts **22***a* and **22***b* not being in contact with each other. Parts **18***a*, **20***a*, **22***a* form a secondary stack **12***a* and parts **18***b*, **20***b* and **22***b* form a secondary stack **12***b*. The secondary stacks **12***a* and **12***b* are separated by the cavity **26**.
[0072] The gate pattern **24** comprises a dielectric layer **28** and a conductive layer **30**. Layer **28** covers conformally the cavity **28** and part of the upper face of the layer **22** surrounding the cavity **26**. In other words, layer **28** covers the lateral face of the layers **16**, **18**, **20**, **22** forming the walls of the cavity, the face of the layer **16** forming the bottom of the cavity **26** and part of the upper face of the layer **22** surrounding the cavity **26**. Layer **30** covers conformally layer **28**. Preferably, layer **30** covers entirely, and only, layer **28**. Layer **30** is therefore separated from stack **12** by layer **28**.
[0073] The transistor **11** comprises a drain electrode **32** and a source electrode **34**. The drain electrode **32** is in contact with both layers **20** and **22** on one side of the cavity **26** and the source electrode **34** is in contact with both layers **20** and **22** on the other side of the cavity **26**. More precisely, the drain electrode **32** is in contact with parts **20***a* and **22***a* of the layers **20** and **22** and the source electrode **34** is in contact with parts **20***b* and **22***b* of the layers **20** and **22**. In the example of FIG. **2**, the drain electrode **32** covers part of the upper face of part **22***a*, part of the upper face of part **20***a* which is not covered by part **22***a*, and the lateral face of part **22***a*. Similarly, the source electrode **34** covers part of the upper face of part **22***b*, part of the upper face of part **20***b* which is not covered by part **22***b*, and the lateral face of part **22***b*.
[0074] The architecture of a HEMT transistor such as transistor **11** includes the superposition of two semiconductive layers having different band gaps which form a quantum well at their interface. This quantum well is induced by the present spontaneous and piezoelectric biasing charges. Electrons are confined in this quantum well to form a two-dimensional electron gas (2DEG). In the stack **12** of FIG. **2**, a two-dimensional electron gas **36** is formed under the layer **22**,

in the layer **20**. The two-dimensional electron gas **36** is typically confined at the interface between the layers **20** and **22**. Layers **20** and **22** form an heterojunction.

[0075] The gate pattern **24** passes through this interface between layers **20** and **22**, so as to interrupt the two-dimensional electron gas **36**. Controlling the gate voltage Vgs makes it possible to enable or block the passage of electrons of the two-dimensional electron gas **36** on either side of the gate pattern **24**.

[0076] Generally, if the gate of the transistor is set to a voltage greater than a threshold voltage, there is an accumulation of electrons under the gate dielectric, thus connecting the two-dimensional electron gas on either side of it, which makes it possible to connect the source and the drain such that the transistor enters an on state. Therefore, when the gate pattern **24** is in a state to enable the passage of electrons, the two-dimensional electron gas **36** extends between the parts **20***a* and **22***a*, along the walls and the bottom of the cavity **26** and between the parts **20***b* and **22***b*.

[0077] If the gate of the transistor is set at a voltage less than the threshold voltage, the source and the drain are no longer connected and the transistor enters an off state.

[0078] It would be advantageous for the threshold voltage to be positive. Indeed, if the value of the threshold voltage is relatively low (<1V), the device can require a negative Vgs voltage to control the off state. The application of a voltage Vgs<0 involves resorting to a more complex control circuit and the increase of the resistance of the transistor to the on-state.

[0079] The layers **16** and **18** allows the increase of the threshold voltage toward higher values.

[0080] However, a two-dimensional electron gas **38** is formed under the layer **18**, in the layer **16**. The two-dimensional electron gas **38** is typically confined at the interface between layers **16** and **18**. The two-dimensional electron gas **38** generates a constant leakage current flowing between the drain and the source of transistor **10**. This current is all the more important in case where the gate pattern does not cross the interface between the layers **16** and **18**.

[0081] Layer **15** is made of p-doped semiconductive material. Therefore, layer **15** is configured to deplete electrons in the neighboring regions. In the embodiment of FIG. **2**, the thickness of layer **16**, in other words, the distance between the two-dimensional electron gas **38** and the layer **15**, is configured to ensure that the two-dimensional electron gas **38** is at least partially suppressed by the influence of the p-type doping of layer **15**. In other words, layer **15** stops the current generated by the two-dimensional electron gas **38**. In other words, layer **15** is configured to deplete the two-dimensional electron gas **38**.

[0082] The choice of the thickness of layer **16** depends on the leakage current generated by the two-dimensional electron gas **38** that needs to be suppressed. In the case of a layer **18** in AlGaN, the leakage current depends on the proportion of aluminum in the layer **18**. Preferably, the thickness of layer **16** allowing the compensation of the leakage current decreases as the proportion of aluminum increases. The thickness of layer **16** also depends on the thickness of layer **18**.

[0083] According to a preferred embodiment, layer **15** is made of GaN doped with magnesium. Preferably, the thickness of layer **15** is higher than 10 nm, preferably higher than 50 nm, for example substantially equal to 100 nm. In the preferred embodiment, layer **15** is floating and does not receive a biasing voltage. According to the preferred embodiment, layer **16** is in intrinsic GaN. Preferably, layer **16** is floating. Preferably, the thickness of layer **16** is lower than 150 nm, preferably lower than 100 nm. According to the preferred embodiment, layer **18** is in AlGaN. Preferably, the proportion of aluminum in layer **18** is higher than 4%, for example substantially equal to 4%. Preferably, layer **18** is floating. Preferably, the thickness of layer **18** is higher than 20 nm, for example substantially equal to 75 nm. According to the preferred embodiment, layer **20** is in intrinsic GaN. Preferably, the thickness of layer **20** is higher than 50 nm, for example substantially equal to 80 nm. According to the preferred embodiment, layer **22** is in AlGaN. Preferably, the proportion of aluminum in layer **22** is higher than 4%, for example substantially equal to 25%. Preferably, the thickness of layer **22** is comprised between 20 nm and 100 nm, for example substantially equal to 24 nm.

[0084] According to another embodiment, the device comprises a contact element configured to bias layer **15**. For example, the device comprises an insulated conductive via, not represented, comprising a conductive core and an insulating sheath. The via for example crosses layers **16**, **18**, **20** to reach the upper face of layer **15**.
[0085] FIG. **3** illustrates the behavior of the electronic device of FIG. **2**. More precisely, FIG. **3** illustrates, in view A, with curves C**1**, C**2**, C**3**, C**4**, C**5**, the conduction band energy, in the different regions of the transistor **11** of FIG. **2** for different values of the thickness of layer **16**. Furthermore, FIG. **3** illustrates, in view A and in view B with curves C**6**, C**7**, C**8** C**9** and C**10**, the concentration of electrons, in the different regions of the transistor **11** of FIG. **2** for different values of the thickness of layer **16**. View B corresponds to a region **41** of view A in larger dimensions.
[0086] In the different cases illustrated as examples in FIG. **3**, layer **15** has a thickness of 100 nm, layer **18** has a thickness of 75 nm, layer **20** has a thickness of 80 nm, and layer **22** has a thickness of 24 nm.
[0087] Curves C**1** and C**6**, C**6** being identical to C**7**, correspond to a case where layer **16** has a thickness of 50 nm. Curves C**2** and C**7** correspond to a case where layer **16** has a thickness of 100 nm. Curves C**3** and C**8** correspond to a case where layer **16** has a thickness of 150 nm. Curves C**4** and C**9** correspond to a case where layer **16** has a thickness of 200 nm. Curves C**5** and C**10** correspond to a case where layer **16** has a thickness of 250 nm.
[0088] The curves of FIG. **3** comprise different parts corresponding to the different regions of the transistor **11**. Part Z**22** corresponds to layer **22**, part Z**20** corresponds to layer **20**, part Z**18** corresponds to layer **18**, part Z**16** corresponds to layer **16** and part **215** corresponds to layer **15**.
[0089] The transistor is considered to be on. All curves C**6** to C**10** comprise a peak at the junction between part Z**22** and Z**20**, corresponding to the two-dimensional electron gas **36** between the electrodes of the transistor.
[0090] In view B, it is possible to see that curves C**8**, C**9** and C**10** comprise a peak at the junction between part Z**16** and Z**18**, corresponding to the two-dimensional electron gas **38** between the electrodes of the transistor. Therefore, the two-dimensional electron gas **38** is not suppressed for the thickness corresponding to curves C**8**, C**9** and C**10**. It is also possible to see that curves C**6** and C**7** do not comprise a peak at the junction between part Z**16** and Z**18**. Therefore, for the corresponding thickness of the layer **16**, the two-dimensional electron gas **38** is suppressed.
[0091] The thicknesses allowing the suppression of the two-dimensional electron gas **38** depend on the composition of the transistor, in particular on the doping values of the different regions and on the thickness of the different regions. It can be determined empirically.
[0092] FIG. **4** illustrates another embodiment of an electronic device comprising a semiconductor-based power transistor **40**.
[0093] The power transistor **40** comprises the elements of the transistor **10**. Those elements will not be described a second time in detail. In other words, the transistor **40** comprises: [0094] the buffer layer **14**; [0095] the p-doped layer **15**, [0096] the piezoelectric layer **16**, [0097] the piezoelectric layer **18**, [0098] the piezoelectric layer **20**, [0099] the piezoelectric layer **22**, [0100] the cavity **26**, [0101] the gate pattern **24**, comprising the dielectric layer **28** and the conductive layer **30**, and [0102] the drain electrode **32**.
[0103] The transistor **40** differs from the transistor **11** of FIG. **2** in that one of the electrodes, for example the source electrode **34** of FIG. **2**, is replaced by an electrode **42**, which is in contact with the layer **15**. For example, a part of the upper face of the layer **14** is not covered by the layers **16**, **18**, **20**, **22**. One of the electrodes, for example the electrode **42**, covers: [0104] part of the upper face of layer **22**, for example the part **22***b*, [0105] the lateral faces of the layers **16**, **18**, **20**, **22**, for example the lateral faces of the parts **18***b*, **20***b*, **22***b* and of the layer **16**, and [0106] the part of the upper face of layer **15** uncovered by the layers **16**, **18**, **20**, **22**.
[0107] In the embodiment of FIG. **4**, layer **15** is biased with the same voltage as one of the electrodes, for example the source electrodes. For example, the layer **15** is biased at the ground. For

example, both the layer **15** and the source electrode **42** is biased at the ground.
[0108] According to an embodiment, the lateral faces of the layers **16**, **18**, **20**, **22** are separated from the electrode **42** by a dielectric layer, not represented. The electrode is then only in contact with the upper faces of layers **15** and **22**.
[0109] An advantage of the described embodiments is that the threshold voltage of the transistor is higher, preferably positive. The transistor is therefore simpler to implement and more reliable.
[0110] Another advantage of the described embodiments is that the energy consumption of the transistor is lower, as a leakage current has been neutralized.
[0111] Another advantage of the described embodiments is that, as the leakage current generated by the two-dimensional electron gas **38** is stopped by the layer **15**, the bottom of the cavity **26** can be above the layer **15** without increasing the leakage current.
[0112] Various embodiments and variants have been described. Those skilled in the art will understand that certain features of these embodiments can be combined and other variants will readily occur to those skilled in the art.
[0113] Finally, the practical implementation of the embodiments and variants described herein is within the capabilities of those skilled in the art based on the functional description provided hereinabove.

Claims

1. A device comprising a transistor, the transistor comprising: a p-doped semiconductor first layer; a piezoelectric semiconductor second layer, covering the p-doped semiconductor first layer; a piezoelectric semiconductor third layer, covering the piezoelectric semiconductor second layer; a piezoelectric semiconductor fourth layer, covering the piezoelectric semiconductor third layer; a piezoelectric semiconductor fifth layer, covering the piezoelectric semiconductor fourth layer, the transistor being configured to generate a first two-dimensional electron gas between the piezoelectric semiconductor fourth layer and the piezoelectric semiconductor fifth layer; and a gate pattern crossing going through the piezoelectric semiconductor fifth layer and at least part of the piezoelectric semiconductor fourth layer.
2. The device of claim 1, wherein the transistor is configured to generate a second two-dimensional electron gas between the piezoelectric semiconductor second layer and the piezoelectric semiconductor third layer, the p-doped semiconductor first layer being configured to suppress the second two-dimensional electron gas.
3. The device of claim 1, wherein the gate pattern crosses the piezoelectric semiconductor third layer, the piezoelectric semiconductor fourth layer, and the piezoelectric semiconductor fifth layer and at least part of the piezoelectric semiconductor second layer.
4. The device of claim 1, wherein the piezoelectric semiconductor second layer and the piezoelectric semiconductor fourth layer are made of the same material.
5. The device of claim 1, wherein the piezoelectric semiconductor second layer and the piezoelectric semiconductor fourth layer are made of GaN.
6. The device of claim 1, wherein the piezoelectric semiconductor third layer and the piezoelectric semiconductor fifth layer are made of the same material.
7. The device of claim 1, wherein the piezoelectric semiconductor third layer and the piezoelectric semiconductor fifth layer are made of AlGaN, AsGa, AlN or InGaN.
8. The device of claim 1, wherein the p-doped semiconductor first layer is made of GaN doped with magnesium, carbon or iron.
9. The device of claim 1, wherein the p-doped semiconductor first layer is configured not to be biased.
10. The device of claim 1, wherein the gate pattern separates at least the piezoelectric semiconductor fifth layer in a first part and a second part.

11. The device of claim 10, wherein the gate pattern separates the piezoelectric semiconductor third layer, the piezoelectric semiconductor fourth layer, and the piezoelectric semiconductor fifth layer in a first part and a second part.

12. The device of claim 1, wherein the device comprises a first electrode in contact with a first part of the piezoelectric semiconductor fifth layer and a second electrode in contact with a second part of the piezoelectric semiconductor fifth layer.

13. The device of claim 1, wherein the p-doped semiconductor first layer is configured to be biased with the same voltage as either the first or second electrode.

14. A method of using the device of claim 1 in on-board chargers in electric vehicles, in charging stations, in photovoltaic systems, in home appliances, in telecommunication systems, in data centers and servers, in light emitting diode lighting systems or in equipment requiring power conversion.

15. A system comprising an on-board charger in electric vehicle, a charging station, a photovoltaic system, a home appliance, a telecommunication system, a data center, a server, a light emitting diode lighting system, or an equipment requiring power conversion, the system comprising the device of claim 1.
